

Numerical Methods

LAB 5

1. Before Eastern we have learned about root finding methods and stopping criteria. The root found by a Newton's method depends on the initial guess, as we have shown for the function $f(x) = x \sin(4x)$ (see Fig. 1):

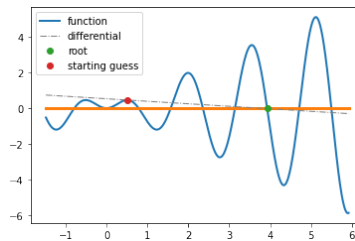


Figure 1: Root finding - starting point dependence.

For this problem (the last task of Lab 4) I made these plots (Fig. 2):

This shows, how the root found by a Newton method depends on the starting point and how the number of iterations depends on it.

Are you able to make these plots? What are your conclusions?

2. Today we have learned about Newton's method for a set of equations. For a set of equations:

$$\begin{aligned}x^2 + y^2 + z^2 &= 2 \\x^2 + y^2 &= 1 \\x^2 &= y\end{aligned}$$

write the equations of the Newton's method (function and Jacobian) for this set of equations.

3. Based on the code from:

Based on code 1.upynb, solve the above set of equations.

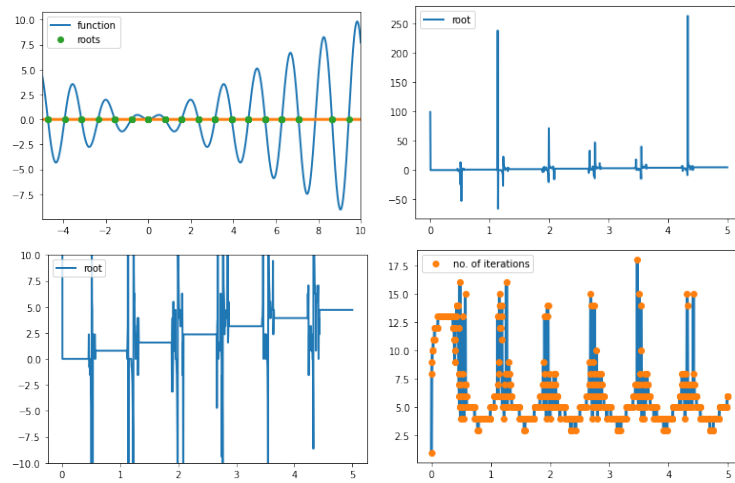


Figure 2: Root finding - dependence on the starting point

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5. EXTRA PROBLEM (not obligatory, just think all of us together):
is there a way to speed the Newton's method in many dimensions using Aitken's acceleration? Try to speed up the "slow" example from the code.